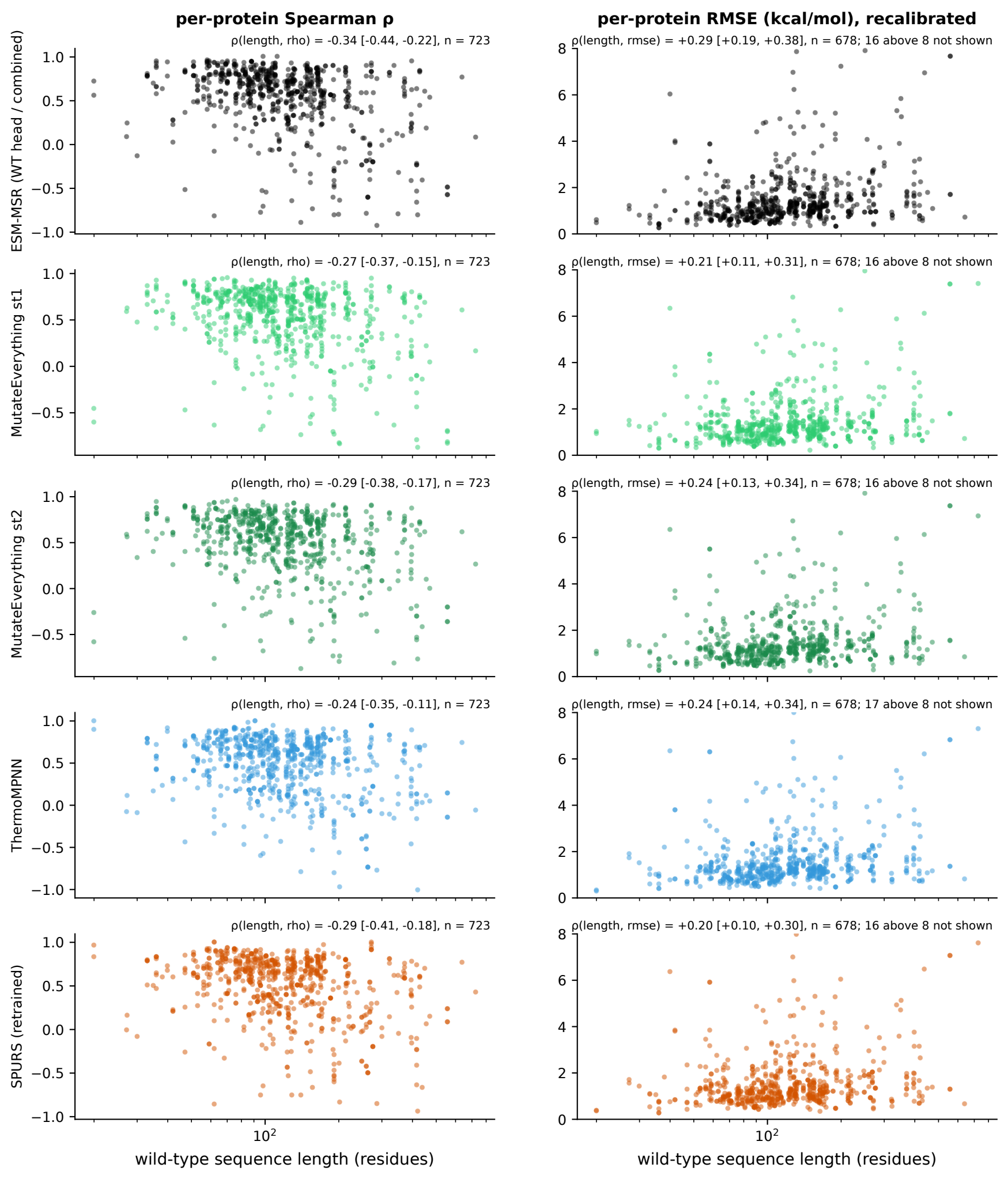

per-protein Spearman ρ
per-protein RMSE (kcal/mol), recalibrated
ESM-MSR (WT head / combined)
MutateEverything st1
MutateEverything st2
ThermoMPNN
SPURS (retrained)
ρ(length, rho) = -0.34 [-0.44, -0.22], n = 723
ρ(length, rho) = -0.27 [-0.37, -0.15], n = 723
ρ(length, rho) = -0.29 [-0.38, -0.17], n = 723
ρ(length, rho) = -0.24 [-0.35, -0.11], n = 723
ρ(length, rho) = -0.29 [-0.41, -0.18], n = 723
ρ(length, rmse) = +0.29 [+0.19, +0.38], n = 678; 16 above 8 not shown
ρ(length, rmse) = +0.21 [+0.11, +0.31], n = 678; 16 above 8 not shown
ρ(length, rmse) = +0.24 [+0.13, +0.34], n = 678; 16 above 8 not shown
ρ(length, rmse) = +0.24 [+0.14, +0.34], n = 678; 17 above 8 not shown
ρ(length, rmse) = +0.20 [+0.10, +0.30], n = 678; 16 above 8 not shown
1.0
0.5
0.0
−0.5
−1.0
8
6
4
2
0
10^2
wild-type sequence length (residues)
wild-type sequence length (residues)